



USER MANUAL / PROTOTYPE EDITION

From maintenance need to a reviewed block plan.

A practical guide for first-time users, planners, officers and administrators.

PLAN	UNDERSTAND	PRESENT
Create weekly and monthly proposals using available access windows.	Read priorities, duration estimates, constraints and source evidence.	Prepare a clear two-minute walkthrough using a saved recommendation.

Application: <https://railblox.vercel.app>

Edition: 1.0 | 12 September 2026

Code reference: 0cb881f

READ THIS FIRST

This manual describes the implemented prototype and its demonstration workflows. Synthetic records and imported evidence support planning proposals. A software approval is not a railway possession authority or permission to dispatch trains. Live railway connectors and measured asset-uptime improvements are not established by this demonstration.

Based on the local application source. Screen labels follow the inspected version; hosted settings and available data can vary. Illustrations and worked examples in this manual are explanatory, not screenshots or measured deployment results.

USER MANUAL / 01

Find your way around

Start with pages 3-5 for your first plan. Use page 12 when preparing a presentation.

Page	Topic	Use it to...
3	Account and roles	Sign in, complete MFA and understand permissions.
4	Your first recommendation	Select a network, review a task and generate a proposal.
5	Read the schedule	Find populated days, inspect packages and validate a plan.
6	Maintenance and source data	Review requests, import files and resolve asset mappings.
7	Opportunities and disruptions	Try compatible additions, changed inputs and benchmarks.
8	Intelligence and AI	Understand rules, statistics, NLP and optional model features.
9	Decisions and reporting	Approve/reject, record actuals and prepare reviewed drafts.
10	Administration and automation	Grant access, manage profiles and configure monitoring.
11	Troubleshooting	Resolve common account, data and planning problems.
12	Two-minute demonstration	Prepare browser tabs and rehearse the presentation.
13	Glossary and reference	Translate planning terms and find implementation sources.

The two navigation rows

Main navigation: Planner, Maintenance, Data review, Opportunities, Disruption lab and Results.

Decision support: Block queue, Operations, Insights and Reporting.

One workflow to remember

INPUTS > RECOMMENDATION > SCHEDULE > REVIEW > OUTCOMES

Start in Block queue. Inspect evidence, generate a recommendation, and use Inspect saved schedule to open that exact proposal. An Officer reviews it; actual execution records can later support reporting and future analysis.

The **Planning network** selector changes the corridor. **Weekly** and **Monthly** select a 7-day or 30-day horizon. Dates and plan versions determine which saved data you are looking at. The planning timezone is Asia/Kolkata (IST).

Account and roles

Use your existing account. Registration, email confirmation, access assignment and MFA are separate steps.

01. Open **railblox.vercel.app**. Enter your email and password and choose **Sign in**.

02. If the app shows pending access, ask an administrator to assign your account to this deployment. A requested role does not grant permissions.

03. If you see **Verify your authenticator**, choose **Set up authenticator**, scan the QR code in your authenticator app, enter its six-digit code and select **Verify and continue**. Existing enrolled users enter a current code.

Role	Main permissions
Viewer	Read workspace records and download exports.
Planner	Viewer access plus maintenance/evidence preparation, recommendations, local analysis and report drafts.
Officer	Planner access plus validation, benchmarks, decisions, task revisions, opportunity additions, finalization and actual outcomes.
Admin	Officer access plus user access management, audit inspection and automation configuration.

New accounts

New user? Create an account appears only when public signup is enabled. Enter the requested profile fields and a role preference. Date of birth is conditional and is hidden by default in production. Complete email confirmation when required, then wait for administrator approval. An invitation also requires a separate access grant.

Email and authenticator codes

Email invitations, confirmations and password resets go to the recipient account email. Resend testing restrictions can prevent delivery to other recipients. Authenticator codes are generated by the enrolled app; they do not arrive by email.

PRODUCTION ACCESS

The inspected code requires MFA for Planner, Officer and Admin in production. Keep the enrollment QR/setup secret private. Lost-factor recovery needs the deployment owner; the app does not provide a self-service recovery-code workflow.

Generate your first recommendation

Recommended starting point: Block queue, with an approved Planner, Officer or Admin account.

01. Choose your demonstration corridor using **Planning network**. Leave it unchanged throughout this walkthrough.
02. Select **Weekly** and the first appropriate date in **Planning start date**. Synthetic data has a fixed date range; the current calendar date may contain no demonstration work.
03. Read the **Block queue**. Pick one task and note its department, priority, duration estimate and review items.
04. Click **Review evidence**. Inspect the priority factors, matched history, dependencies and window conflicts. Close the panel when finished.
05. Click **Recommend block plan** once. Wait for the job to finish. The app saves the recommendation automatically.
06. Check the result heading: **X work items in Y block packages**. For a demonstration, both numbers should be greater than zero. Look for assigned times under **Recommended work interval**.

Read the queue correctly

Field	Meaning
Priority / score	A weighted policy assessment. A score range indicates missing assessment values. Mandatory work is kept as a modeled requirement.
Duration estimate	An empirical range from similar completed work, where enough history exists. It is not a guaranteed completion time.
Recommended work interval	The proposed window and start/end time for an assigned task. Unassigned work needs review.
Review items	Issues such as unresolved locations, readiness or insufficient history. Inspect the source evidence.

EXPECTED RESULT

The recommendation should be current and have a feasible result before approval. If it says No feasible recommendation to approve, inspect constraints and data. Do not remove modeled protection requirements simply to obtain a successful demo.

Use **Saved recommendation** to revisit a stored result. Saving a priority assessment or importing new evidence can make an earlier recommendation outdated; regenerate it before a new decision.

Read and validate the schedule

Open the correct saved plan before interpreting its timeline.

01. In the recommendation, click **Inspect saved schedule**. This opens Planner with the associated plan ID.
02. Scroll to **Weekly schedule** or **Monthly schedule**. Each day button shows a package count. Click a day with at least one package.
03. Read the timeline. Inspect work bars or packages to see tasks, intervals and related details. Use **Reset filters** if work is hidden by a department, section, priority or status filter.
04. Review **Unscheduled work** and its reasons. An omitted task is not proof it was completed or unnecessary.
05. As Officer or Admin, select **Validate**. Review the check results against the selected plan and input version.

Two planning entry points

Block queue > Recommend block plan applies the evidence-based priority and historical duration allowances before solving. **Planner > Generate plan** solves the selected scheduling snapshot. To demonstrate a recommendation, use its **Inspect saved schedule** link rather than showing an unrelated plan version.

Indicator	How to interpret it
FEASIBLE	A valid solution was found for the modeled constraints; global optimality is not established.
OPTIMAL	The solver proved optimality for the encoded model/objective, not every real-world railway consideration.
Current / inputs changed	Shows whether the proposal still matches the source state. Refresh and regenerate outdated work.
Closed-section-minutes	Closure time summed across affected directed sections. It is a planning metric, not measured asset uptime.

DATES AND VERSIONS

The date dropdown changes the planning start; the day buttons change which day is visible. Changing Weekly/Monthly or the start date does not rewrite an existing plan. Generate a new plan when applying a new horizon.

Compare versions compares saved results. **Export JSON** and **CSV** download records. **Approve proposal** and **Finalize cycle & record deferrals** change workflow state; use them only after review, not simply to navigate.

Maintain inputs and evidence

Use **Maintenance** for work requests, **Data review** for maintenance imports, and **Operations** for contextual evidence.

Maintenance requests

Open **Maintenance** or the queue link **Maintenance requests**. Search/filter work and open a task to inspect its location, duration, department, resources and status. Planner-level accounts can prepare inputs; saved changes can invalidate existing proposals. Resolve location and readiness issues before planning.

Import maintenance data

01. Open **Data review**. Download a **Sample CSV** or **JSON** from the relevant source tile. Use its structure as the illustrative import contract.
02. Choose the matching **Source**, select your CSV/JSON file and click **Preview records**.
03. Read errors and mapping requirements. Check that the columns and chainage units match the intended fields. Confirm the displayed mapping, then choose **Commit valid records**.
04. In **Location mapping review**, choose a verified candidate location for unresolved items and click **Confirm mapping**. Generate a fresh proposal from the updated source snapshot.

Import operational context

01. Open **Operations** and inspect the source-status cards and train occupancy inputs. These represent supplied evidence, not live connections.
02. Use **Download evidence JSON**, then prepare a compatible file containing the current snapshot reference, feed timestamps and relevant blocks, cautions, weather and history.
03. Choose **Import evidence JSON**, review the preview and select **Validate and import evidence**. New evidence supersedes earlier recommendations.

KNOW THE DATA ORIGIN

The source names TMS, SMMS, TDMS, COA and others identify prototype adapters or evidence roles. The app does not contain authenticated live railway connectors. Keep synthetic fixtures labeled synthetic; imported records need their real provenance and timestamps.

Stale, missing, unavailable or future-dated feeds can block recommendations. Refresh with valid source records. In the synthetic demo, freshness uses the displayed evidence clock.

Explore alternatives

Use saved baseline plans so each trial has a clear before-and-after reference.

Find compatible work in an existing package

01. Open **Opportunities** and choose a **Proposed package**. Select **Discover compatible work**.
02. Inspect compatible/rejected bundles. Read the added closed-section-minutes, resources and remaining margin.
03. An Officer or Admin can choose **Add to proposed plan** for eligible work. This creates a revised proposal; recheck its result.

The search tries candidate additions while preserving existing assignments. A zero increase in closure can still use a scarce crew or reduce the available margin. Compatibility is a solved constraint result, not just matching task names.

Test changed inputs

01. Open **Disruption lab**. Choose the scenario appropriate to your test, such as a shorter window, freight change or crew unavailability.
02. Select the target and enter the requested time values. Click **Create scenario snapshot**.
03. Read **Changed inputs**, then click **Repair plan**. Inspect changed assignments, start shifts, closure footprint and validation results.
04. Use **Compare before / after**. **Reset to baseline** returns the displayed selection to the starting plan; it does not erase saved history.

Compare planning methods

In **Results**, Officer/Admin users can choose **Run four methods**. The comparison covers sequential departmental greedy planning, joint greedy planning, deterministic CP-SAT and adaptive repair. Read measured coverage, closure and scenario outcomes together. Unequal service coverage matters when comparing closure cost.

DEMONSTRATION TIP

Alternative planning is useful for a longer demo. For a two-minute presentation, keep one saved baseline and one clear package. Avoid spending the limited time waiting for repeated scenario or benchmark jobs.

Understand the intelligence

Local means computation within the RailBLOX backend, including its Render deployment. It does not mean a model running on the viewer's laptop.

Feature	Where to find it	What it actually does
Priority policy	Block queue > Review evidence	Uses explicit weighted factors. It is not a learned failure probability.
Historical durations	Queue and Insights	Matches actual/planned duration ratios; shows median and P20/P80 estimates. The recommendation uses a conservative allowance.
Local narrative analysis	Insights > Delay and failure narrative analysis	Finds vocabulary matches, exact text spans, known sections, durations and simple negation.
Duration regression	Insights > AI services and learning	Fits ridge regression using NumPy and evaluates it on later completed records. Evaluation-only.
Scheduling optimizer	Recommend block plan / Generate plan	OR-Tools CP-SAT searches assignments under modeled constraints.
Optional LLM	AI explanation / report summary / incident analysis	Requests structured draft text from the configured OpenAI model. Requires backend configuration.

Try local narrative analysis

Under **Delay and failure narrative analysis**, enter: *Signal failure near S1-UP. Work took 45 minutes. No OHE fault.* Replace S1-UP with a section that exists in your workspace. Click **Analyze narrative**. Inspect the signal-failure match, duration and negated OHE-fault mention. Severity and causality still require review.

Try the learning experiment

Select **Train and evaluate duration model**. Training requires at least 30 completed outcomes, including at least 20 earlier training and 10 later holdout records. Compare model MAE with the engineering baseline; smaller MAE means lower average error on that holdout. The experiment does not replace scheduling allowances automatically.

ACCURATE AI CLAIM

The built-in analysis combines rules, statistics and optimization. PyTorch, TensorFlow and a local LLM are not used. Optional OpenAI calls run only when explicitly requested; they do not approve or dispatch work. Synthetic evaluation is not evidence of real railway prediction accuracy.

Review, record and report

Recommended workflow: keep the decision attached to the same saved recommendation and evidence.

Approve or reject a recommendation

01. In **Block queue**, choose the intended **Saved recommendation**. Check its current status, feasibility, evidence and assigned/unassigned work.

02. As Officer or Admin, enter an **Officer review reason** describing your assessment.

03. Choose **Approve recommendation** or **Reject recommendation**. Approval rechecks the plan against modeled constraints. A decided recommendation is not simply overwritten.

To request an alternative before decision, expand **Modify a recommended window**, select work and a window, provide the review reason and choose **Recompute modification**. The new solve remains a proposal. Started or locked work needs the explicit commitment-revision workflow.

Record what actually happened

Use **Record actual execution** for a task assigned to an approved recommendation. Enter actual start/end, observation time, delay, observed weather and narrative; select **Save actual outcome**. These inputs use minutes from the displayed anchor. For example, 90 means anchor time plus 90 minutes. Do not enter a clock value such as 0930 unless it really is that offset.

Prepare a report

01. Open **Reporting**. Verify the saved recommendation reference. Choose **Draft type** (MIS or PAM), then **Prepare report draft**.

02. Check missing actuals and source fields. Missing records remain missing; a planned time is not an actual completion.

03. An Officer/Admin edits the summary, adds a **Report review reason** and clicks **Record draft review**. Use **Export draft JSON** to download it.

WHAT APPROVAL MEANS HERE

App approval records a prototype workflow decision. Report exports use illustrative MIS/PAM fields and do not submit official returns. Prepare a new report draft after additional outcomes; existing drafts retain their captured data.

Administration and monitoring

Admin users manage access and schedules for automation. The same application serves all roles; controls and backend permissions differ.

Grant access without SQL

01. Open **Your profile**, then **Manage user access**. In **Access review**, locate the colleague. The filter searches the current page; use **Next accounts** when needed.

02. Choose **Review access**. Check email confirmation, current assignment and the target division.

03. Select **Assigned role**, enter an **Access change reason**, and click **Confirm access assignment**. A signup role is only a preference.

04. Use **Refresh access records** and inspect **Audit history**. The app rechecks identity periodically and on tab focus; backend requests verify access.

Invite a colleague sends an actual email through Supabase SMTP. Confirm the recipient before using **Send invitation email**. Invitations do not grant access. If Resend is limited to its test sender, other recipients cannot complete email delivery. The admin API also needs backend service-role configuration.

Your profile

Review your account email, division, role and session verification. Edit **Full name** and select **Save profile**. Use **Change password** for the password flow. When available, **Profile privacy** lets you remove a previously stored date of birth. A user cannot self-assign a role.

Enable monitoring and report preparation

01. In **Insights**, find **Scheduled monitoring and report preparation**.

02. Set an interval from 5 to 1440 minutes. Select **Enable schedule** and, if needed, **Prepare report drafts**. Click **Save automation settings**.

03. Use **Run monitoring now** to check immediately. Read the run status, alert count and newly created drafts.

AUTOMATION SCOPE

Monitoring checks stored evidence and creates MIS/PAM drafts for current approved recommendations. Identical recommendation/outcome sets do not create duplicate automated drafts. It does not call the LLM, fetch live railway feeds or approve work. Scheduled runs require an active backend process; hosting sleep or downtime interrupts execution.

When something does not work

Start with the message shown by the app. Keep the request ID if one is displayed.

Symptom	What to do
No Create an account option	Ask the deployment owner whether self-registration is enabled. Both RailBLOX and Supabase signup settings apply.
Confirmation/reset email missing	Check the recipient inbox/spam and provider logs. Resend's test sender can restrict delivery to the account owner. Deployment alone does not remove that restriction.
An administrator must grant access	A confirmed account still needs an assigned role for this deployment division. Ask Admin to review access.
Verify your authenticator	Complete TOTP enrollment or enter a current code. For a lost factor, use the deployment owner's recovery process.
Recommend block plan disabled	Check Planner-or-higher access, whether a job is already running, and the source feed status in Operations.
No feasible recommendation	Inspect date/horizon, mandatory requirements, task readiness, mappings, resources and occupied intervals. Resolve input problems and recompute.
Empty timeline	Use Inspect saved schedule, click a day with a nonzero package count and Reset filters. Confirm the selected version.
Evidence changed / stale snapshot	Refresh and recompute from current inputs. Earlier saved results remain useful as history.
Insufficient duration history	The app falls back to engineering estimates. Regression needs the minimum completed training/holdout records.
LLM assistance disabled	Local planning remains available. The optional provider needs a key, model and permitted data policy configured on the backend.
Planning service unavailable	Check the backend deployment and connection configuration. A loading/waking service is different from an invalid plan. Retry after the service is available.

BEFORE SHARING A SCREENSHOT

Share the visible error and request ID. Exclude passwords, API keys, full database URLs, authenticator QR codes and setup secrets. For presentation backups, capture only planning screens.

Your two-minute demo

Tell one story: a maintenance need becomes a coordinated, reviewable schedule.

Prepare the three browser tabs

- 01.** Sign in and complete MFA beforehand. In **Block queue**, choose the demonstration network/date and **Weekly**. Generate a recommendation; confirm it schedules work.
- 02.** Right-click **Inspect saved schedule** and open it in a new tab. In its calendar, choose a day showing at least one package.
- 03.** Duplicate the original queue browser tab. In the duplicate, click **Insights** and leave the intelligence brief visible. Keep the same network across all tabs.
- 04.** Capture each planning screen with **Windows + Shift + S**, open the capture and save it as 01-queue, 02-schedule and 03-insights. Keep them ready as clearly identified backup captures.

Time	Screen and spoken script
0:00-0:20	Block queue. "RailBLOX helps coordinate railway maintenance within available access windows. This demo uses synthetic data."
0:20-0:40	One task / Review evidence. "Here are the task's priority factors and duration estimate. We can trace the recommendation back to its inputs and similar completed work."
0:40-1:25	Saved schedule. "OR-Tools creates a schedule respecting modeled windows, resources and train occupancy. This package contains [read the actual tasks shown]. Compatible work can share access."
1:25-1:45	Insights. "The app highlights historical duration overruns and evidence needing review. These local rules and statistics require no external AI key."
1:45-2:00	Recommendation review. "An officer reviews the proposal before approval. This prototype demonstrates coordinated planning; live integration and measured uptime improvement are the next validation steps."

REHEARSE ONCE

Use a phone timer. Show a saved result if live generation is slow, and say it is saved. Read only the numbers actually visible. Do not claim a downtime-reduction percentage without a measured, comparable result.

Glossary and reference

These definitions explain the prototype interface. They are not a replacement for an operator's railway rulebook.

Term	Plain-language meaning
Block / possession	A reserved maintenance-access period with modeled restrictions on affected infrastructure or train movements. Actual authority follows railway procedures.
Corridor / directed section	The selected planning network and a section identified by its direction or line.
Window / package	A window is available access time; a package groups compatible scheduled work within it.
ENG / TRD / S&T	Engineering, traction distribution, and signalling & telecommunications department labels used in the app.
Mandatory / optional	A modeled requirement versus work that can remain unscheduled when constraints or priorities prevent selection.
Anchor / horizon	The reference timestamp for minute offsets; the 7-day or 30-day planning period.
Snapshot / evidence / version	Saved planning inputs, supporting records and the identifiable stored state used for a result.
P20 / P80 / MAE	Empirical duration percentiles; mean absolute error measures average prediction error in minutes.
AAL2 / MFA	A session that has completed a second authentication factor; required for operational roles in production.

What this edition is based on

Reviewed against commit **0cb881f**. Interface references: WorkspaceShell, App, DecisionWorkspace, AITools, AccountBoundary, MfaGate, AdminWorkspace and ProfilePage. Backend references: security.py, intelligence.py, intelligence_api.py, learning.py, solver.py and platform_api.py.

Companion documents in **docs/**: deployment.md, spec2-implementation.md, spec3-implementation.md and phase-2-asset-availability.md. The phase-2 document describes future work, not delivered features.

Repository: github.com/popeye05/RailBlox | Application: railblox.vercel.app

PROTOTYPE BOUNDARY

The implementation supports advisory planning, user roles, evidence review and stored analysis. It does not establish live railway integration, validated accident-risk prediction, guaranteed uninterrupted train operation or measured asset availability gains.